LEAKLISTEN · DESIGN PRECIS

LeakListen design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Daily charge, battery life and cost updated for the status light pipe; general arrangement Rev P5; no requirement status changed
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02: R12 restated (met on paper), calibration check in CalRig, first candidate partner region
0.5	2026-10-01	Amish Chadha	Constructable design (LKL-DDR-003): neck bar hanger in place of the strap, flanged end plugs, eye bolt and SMA bulkhead, internal chassis, 8 mm puck base; numbers from LKL-CAL-001 v0.3; LKL-DWG-001 Rev P4; cost against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002): 42 mm pot magnet with keeper plate; contact sensor scoped to metallic mains, hydrophone variant for plastic; district gateway within 0.5 km under iron covers; numbers from LKL-CAL-001 v0.2; LKL-DWG-001 Rev P2

0.3	2026-09-25	Amish Chadha	TRL 3 design choices adopted for TRL 3 (LKL-DDR-001); numbers replaced by LKL-CAL-001; 53 g seismic mass, 24-byte summary, regional transmit power, weekly time correction; links to the model and LKL-DWG-001
0.2	2026-09-25	Amish Chadha	TRL 2 concept, components, first-order numbers, safety and open questions
0.1	2026-09-25	Amish Chadha	Initial scaffold

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Summary

LeakListen is a battery-powered acoustic leak logger that hangs under a valve chamber cover, sticks a magnetic vibration sensor onto the valve spindle cap, listens to the pipe for a few minutes each night, and sends one small LoRaWAN summary a day. A leak on a pressurized main makes a steady hiss that travels along the pipe wall; when a logger's quietest night-time level rises and stays up for several nights, the server flags that street for a follow-up survey. The TRL 3 calculations (LKL-CAL-001 v0.3) give about 1.2 mAh a day, 10 years or more on one C-size lithium cell, 1.35 kg with the neck bar hanger and about $142 in parts for the constructable design (LKL-DDR-003). They also show the two weak points, now answered by decisions Amish accepted on 2026-09-25 (LKL-DDR-002). The reference leak is heard about 185 m along an iron main but only a few metres along a plastic one, so the magnet-on sensor is scoped to metallic mains and plastic networks are left to a hydrophone variant (R2, not yet sized). The radio reaches about 0.5 km, not 1 km, from under a cast-iron cover, so a gateway is planned within 0.5 km of each such district (R7, at risk until the cover loss is measured).

Figure 1. Concept in a valve chamber, street and chamber shown in section (media/hero.png). Figure 2. Exploded view with BOM numbers (media/exploded.png). Figure 3. Cutaway of the logger and sensor (media/cutaway.png). Figure 4. Nightly data flow (media/flow.png). Figure 5. General arrangement, drawing LKL-DWG-001 Rev P4 (cad/drawings/LKL-DWG-001.pdf), from the parametric model cad/src/model.py.

How it works

1. **Place from the surface.** A technician lifts the cover, lowers the sensor puck on its cable with a pole until the magnet snaps onto the valve spindle cap, and sets the neck bar, with the logger and antenna already fitted, across the chamber neck just below the cover frame, winding out one rubber foot by hand; the logger hangs from the bar on a lanyard and the flat antenna sits on the bar 20 mm under the cover. Nobody enters the chamber. (The concept hooked a strap over the frame; that is not buildable without lifting the cover off its seat, LKL-DDR-003.)
2. **Listen at night.** A real-time clock wakes the logger between 02:00 and 04:00, when demand and traffic noise are lowest. It records 12 windows of 20 s at 8 kS/s from the piezo sensor through a charge preamplifier and a 24-bit ADC. Once a week it asks the network for the time (LoRaWAN 1.0.3 DeviceTimeReq) so the clock stays within 20 s.
3. **Process on the device.** For each window the microcontroller computes the RMS level and a 64-band spectrum from 5 Hz to 2 kHz, in fixed point, with three FFT stages at 8,000, 1,000 and 250 S/s so that even the 0.49 Hz wide bottom band is resolved. It keeps the minimum and the 10th percentile level across the night (leaks are steady; traffic and water use are not) and a band spectrum. Raw samples are then deleted.
4. **Send one summary.** 24 bytes (two night levels, 16 band levels, peak band, steadiness, battery voltage, temperature, flags) go by LoRaWAN to any network server, such as the lab's TwinKit gateway, at +14 dBm in EU868 or up to +20 dBm in US915. The frame fits every EU868 rate and US915 at SF9 or faster.
5. **Flag on the server.** After a 14-night baseline, a logger whose night-time minimum rises by a set margin and stays up for 3 or more nights, with a steady narrow-band spectrum, is flagged. A crew then pinpoints with a ground microphone or a correlator.

Main components

Numbers match the exploded view and bom/bom.csv.

Table 1. Main components.

No.	Component	Key figures	Role
1	Sensor puck body	Aluminium, 40 mm diameter x 42 mm, potted	Stiff, sealed carrier for the sensor and preamplifier
2	Pot magnet	42 mm, about 600 N nominal pull on thick flat steel; keeper plate for transport (LKL-DDR-002; was 32 mm, 290 N)	Attaches to the spindle cap with no tools; 125 N on a 0.5 mm coating (R9)
3	Piezo disc and seismic mass	27 mm brass-backed piezo disc in compression under a 20 x 20 mm brass mass, about 53 g; about 157 pC/g	Turns pipe vibration into charge
4	Charge preamplifier	Low-noise JFET-input stage, 1 nF and 53 MΩ feedback (3 Hz high-pass), then 40 dB; about 1.0 µg/√Hz	Low-noise, low-frequency front end (R3, R4)
5	Sensor cable	2 m shielded PUR, M12 IP68 plug	Lets the puck sit on the valve while the logger stays near the cover
6	Logger housing	63 mm PVC tube, 240 mm flange to flange, two flanged acetal end plugs with two O-rings each and three radial screws, M12 panel socket	IP68 enclosure (R11)
7	Main board	STM32WL LoRaWAN module (FieldNode core), 24-bit audio ADC, RTC, hybrid layer capacitor of 0.1 F or more	Scheduling, spectrum, radio
8	Primary cell	Li-SOCl2, C size, 3.6 V, about 7.7 Ah	Years of life without charging (R6)
9	Neck bar hanger	Telescopic 20 and 16 mm square aluminium tube with two rubber levelling feet, wedged across the chamber neck below the frame; wire rope lanyard and snap hook	Surface installation (R10) without disturbing the cover
10	Flat LoRa antenna	About 70 mm disc on a bracket on the neck bar, 20 mm under the cover; 1 m lead with an SMA plug	Radio as close to the cover as possible (R7)
12	Internal chassis	Printed PETG spine hung from the top plug, clips for the cell and capacitor, desiccant pocket	Holds the electronics; lifts out with the top plug
13	Top plug fittings	M6 eye bolt (lanyard) and IP67 SMA bulkhead (antenna); the status light pipe rod, part of line 7, shows a short blink at power-up or magnet swipe	Each sealed on its own washer or O-ring

Key design choices

These choices were proposed at TRL 2 and were decided by Amish on 2026-09-25: go with recommendation (LKL-DDR-001, confirmed in LKL-DDR-002). The last three follow from the TRL 3 review, also decided by Amish on 2026-09-25 (LKL-DDR-002).

- **Contact sensor on the valve, not a hydrophone.** A magnet-on sensor needs no pipe tapping and no contact with drinking water. Hydrophones hear further on plastic pipes but need a hydrant or tapping point; this is kept as an option for plastic networks (D3), and LKL-CAL-001 shows it is needed to hear any distance on PVC or PE.
- **Noise-level screening, not correlation.** Correlation needs time-synchronized raw audio from pairs of loggers and much more data over the radio. Nightly level and spectrum screening fits LoRaWAN and a primary cell (D1); correlation is recorded as a later variant.
- **Primary cell, not solar.** A chamber has no sun, and a lithium thionyl chloride cell has very low self-discharge. A hybrid layer capacitor supplies the radio pulses the cell cannot deliver alone. A C-size cell (D5) gives 10 years or more.
- **Reuse the FieldNode radio core.** The STM32WL module, payload conventions and network settings are shared with FieldNode so fixes carry across; FieldNode's solar charger and enclosure are not used (D4). Transmit power and the link budget method follow FieldNode's FND-CAL-001.
- **Compression-mode sensor with a heavy mass.** A 53 g brass mass on the disc in compression gives five times the TRL 2 sensitivity while keeping the sensor's own resonance near 200 kHz, far above the band. A disc used as a bender would be as sensitive but would resonate inside the band.
- **Privacy by design.** Raw samples never leave the logger (R8, a fixed requirement under D6). A contact sensor on a valve is a poor microphone, but the rule keeps the design acceptable for street deployment.
- **Contact sensor for metallic mains, hydrophone variant for plastic.** LKL-CAL-001 shows a contact sensor hears only a few metres along PVC or PE, so the magnet-on sensor's scope is metallic mains (R1), and R2 applies to the hydrophone variant on a hydrant or tapping point, which is not yet sized.
- **District gateway under iron covers.** The flat antenna stays under the cover. Where covers are cast iron, a gateway (TwinKit or a partner's) is planned within about 0.5 km of the district; a through-cover antenna or composite cover is used only where the utility agrees. The cover loss is to be measured before deployment (TRL 4, on hold).
- **42 mm pot magnet.** About 600 N rated, it holds 125 N on a cap with 0.5 mm of paint and rust, against 60 N for the 32 mm magnet, for $4 more. It ships with a keeper plate fitted.

Key numbers (LKL-CAL-001)

All values come from LKL-CAL-001 and `docs/04-calcs/sizing.py`; they are first-principles estimates on assumed data.

Table 2. Key numbers.

Quantity	Value	Requirement
Data sampled and sent per night	3.84 MB sampled and deleted; 24 bytes sent	R8, R14
Daily charge	1.18 mAh (EU868 SF7) to 1.25 mAh (EU868 SF12), with the status light blink	R6
Battery life on energy	10.1 to 12.5 years; taken as 10 years	R6 (5 years)
Airtime	1.97 s per uplink at SF12; 5.9 s a day worst case	R14
Link under an iron cover (20 dB assumed)	About 0.53 km range (EU868 SF12); +0.9 dB at the planned 0.5 km gateway distance, -1.2 dB for US915 SF9	R7 at risk
Detection distance, reference leak	About 185 m on ductile iron (93 to 370 m); about 4 m on PVC with the contact sensor, out of its scope	R1 at risk; R2 open (hydrophone variant)
Sensor self-noise	1.03 µg/√Hz worst from 100 Hz to 1 kHz	R4 at risk
Mounted resonance	About 676 to 1,350 Hz with the 42 mm magnet and 8 mm puck base	R3 at risk
Magnet hold on iron, 42 mm magnet	309 N bare, 187 N at 0.3 mm, 125 N at a 0.5 mm coating	R9 met on paper
Clock	20 s worst with a weekly time correction	R5
Size and mass	63 mm tube (68 mm over the screw heads), 283 mm overall; 1.35 kg with the neck bar, 0.96 kg for the logger, cable and sensor	R12 met on paper: 0.96 kg against 1.0 kg for the logger, cable and sensor (restated 2026-10-02, LKL-DDR-003 A1); the bar is site hardware
Parts cost	About $142 against a value-engineering target of $130	R15: USD 12 over the target

Energy. Listening, 12 mA for about 252 s a night, is 59 % of the worst day; sleep at 4 µA and the cell's own self-discharge make up most of the rest. The radio adds only 0.005 to 0.08 mAh a day at +14 dBm.

Radio. The flat antenna under the cover is the weak link. To close 1 km, the cover may cost no more than about 10 dB. Decided by Amish, 2026-09-25 (LKL-DDR-002): measure the cover loss first, then plan a gateway within about 0.5 km of each district with iron covers, with a through-cover antenna or composite cover where a utility agrees. At 0.5 km the EU868 link has +0.9 dB to spare at a 20 dB cover loss, and the US915 SF9 link misses by 1.2 dB.

Cost. Value-engineering target: USD 130. Estimated cost of the constructable design: USD 142 (USD 12 over the target), see bom/bom.csv. The neck bar, the turned end plugs, the chassis and the top plug fittings added USD 28 to the concept's USD 113.

Safety

Safety: Valve chambers can be confined spaces with low oxygen or toxic gas. LeakListen is designed to be placed from the surface; nobody should enter a chamber without the utility's confined space permit, gas testing and a trained standby person. Covers can weigh 30 kg or more: use a lifting key, lift with the legs and keep feet clear. Street work needs traffic management and high-visibility clothing under local rules.

The lithium thionyl chloride primary cell can vent, burn or explode if shorted, crushed, heated above its rating or charged. Fit a fuse or PTC and reverse protection, never fit a charging circuit, and store and ship cells under the dangerous goods rules for lithium metal cells.

The 42 mm pot magnet pulls about twice as hard as the 32 mm one it replaces, and can pinch fingers badly and affect pacemakers and implanted devices. Carry it with its keeper plate fitted and remove the plate only at the cap.

Placing anything on a public water network needs the utility's written permission. Nothing in LeakListen touches drinking water.

Open questions

- [ ] Detection distance on iron pipes (R1): estimated in LKL-CAL-001, but the leak source strength and pipe losses need field recordings of known leaks.
- [ ] Sizing the hydrophone variant for plastic networks (R2).
- [ ] Radio loss through cast-iron, ductile iron and composite covers (R7): assumed 10 to 30 dB in LKL-CAL-001, not measured. The response is decided (district gateway within 0.5 km by default, LKL-DDR-002); the measurement is TRL 4 work and on hold.
- [ ] Sensor self-noise (R4): 1.03 µg/√Hz on assumed data; the ceramic's d33 and the op-amp noise need checking against chosen parts.
- [ ] Threshold rules for the leak flag, and how to handle pumps, pressure reducing valves and night-time customer use (R13).
- [ ] Adapter for brass, bronze and plastic fittings (the 42 mm magnet for coated caps is decided, LKL-DDR-002).
- [ ] Whether time synchronization for correlation between neighboring loggers is worth adding later.
- [x] Logger calibration check: a small shaker bay in CalRig, which today covers temperature, humidity and particles only; LeakListen defines the acceptance test (frequency range, level and pass band) in its own repo (decided by Amish, 2026-10-02).
- [x] First partner: a water utility with metallic mains, district metered areas and a leakage target it reports against; the first candidate region to approach is the UK, which would set the radio band to EU868 (decided by Amish, 2026-10-02; nothing agreed). The pilot utility is asked for its range of chamber neck sizes before any deployment.